Matter in Our Surroundings

The big idea

Everything that has **mass** and occupies **space** is matter: air, water, your desk, even the smell of food. Matter is made of **tiny particles**, and how those particles behave explains the three states and how they change.

Characteristics of particles

PARTICLES OF MATTER

1. They are **extremely small**.
2. They have **space between them** (sugar dissolves in water without raising the level much).
3. They are **continuously moving** (kinetic energy increases with temperature).
4. They **attract each other** (it's harder to break iron than chalk).

Diffusion is the intermixing of particles of two substances on their own. It is fastest in gases, slower in liquids, very slow in solids, and faster at higher temperature (a drop of ink spreads faster in hot water).

States of matter

Property	Solid	Liquid	Gas
Shape and volume	Fixed shape and volume	Fixed volume, takes container's shape	No fixed shape or volume

Property	Solid	Liquid	Gas
Compressibility	Negligible	Very little	Highly compressible (LPG, CNG cylinders)
Interparticle space	Least	More	Most
Force of attraction	Strongest	Weaker	Weakest
Rigidity / fluidity	Rigid	Flows	Flows

Change of state

- **Melting point**: temperature at which a solid melts at atmospheric pressure. Ice: **0 °C (273 K)**.
- **Boiling point**: water boils at **100 °C (373 K)**.
- **Latent heat**: heat supplied during a change of state **doesn't raise the temperature**; it is used to overcome the attraction between particles. Latent heat of fusion (melting), of vaporisation (boiling).
- **Sublimation**: solid directly to gas without melting: camphor, naphthalene, ammonium chloride, dry ice (solid CO_2). Gas directly to solid is **deposition**.
- **Pressure** also changes state: increasing pressure and lowering temperature can liquefy gases.

KELVIN AND CELSIUS

$$T(K) = T(°C) + 273$$

Evaporation

Evaporation is a surface phenomenon: particles at the surface escape as vapour **below** the boiling point. It increases with **surface area**, **temperature** and **wind**

speed, and decreases with **humidity**.

EVAPORATION CAUSES COOLING

Escaping particles take energy with them, so the remaining liquid cools. That's why sweating cools us, an earthen pot (matka) keeps water cool, and cotton clothes are best in summer (they absorb sweat and let it evaporate).

Quick recap

- Matter is particles with space, motion and attraction.
- Solid, liquid, gas differ in spacing and attraction.
- Latent heat is absorbed at constant temperature during a change of state.
- Evaporation is a surface effect that causes cooling.

Practice questions

1. Which state of matter is most compressible?
   (a) Solid (b) Liquid (c) Gas (d) All equally
2. 25 °C in kelvin is
   (a) 248 K (b) 298 K (c) 273 K (d) 25 K
3. Which substance sublimes?
   (a) Common salt (b) Sugar (c) Camphor (d) Ice
4. Evaporation increases when
   (a) humidity increases (b) wind speed increases (c) surface area decreases (d) temperature decreases
5. During melting, the temperature stays constant because the heat
   (a) is used to overcome forces of attraction between particles (b) is lost to the surroundings (c) makes particles bigger (d) is converted to light

Answers: 1. c 2. b 3. c 4. b 5. a